UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Short notes and keywords

Short answers, a keyword index, rapid recall and a last-10-minutes page.

Class 006 · Particular integral Type 4 ($e^{ax}V$) and Type 5 (xV) · compiled 16 October 2026

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1 Short notes

1 · The Type 4 rule: $X = e^{ax}V$ (exponential shift)

Type 4: $X = e^{ax}V$. $\frac{1}{f(D)}(e^{ax}V) = e^{ax}\frac{1}{f(D+a)}V$: take e^{ax} to the left, replace every D by $D+a$, simplify, then use the rule for V ($\sin/\cos$: $D^2 \to -b^2$). Reason: $D(e^{ax}V) = e^{ax}(D+a)V$. Examples: $\frac{e^x\cos x}{D^2-2D+4} = e^x\frac{\cos x}{D^2+3} = \frac{e^x\cos x}{2}$; $\frac{e^x\sin x}{(D-1)^2} = e^x\frac{\sin x}{D^2} = -e^x\sin x$; CW P17: $\frac{e^{2x}\cos x}{D^2+4D+3} = \frac{e^{2x}(7\cos x+4\sin x)}{130}$. A root a of the AE gives a factor D after the shift.

2 · Type 4 with a polynomial: $x^m e^{ax}$ and mixed right-hand sides

V a polynomial: $\frac{x^m e^{ax}}{f(D)} = e^{ax}[f(D+a)]^{-1}x^m$, take $f(a)$ out, expand up to D^m. $\frac{x^2e^{3x}}{D^2+2} = \frac{e^{3x}}{11}(x^2 - \frac{12x}{11} + \frac{50}{121})$. If $f(D+a) = D^k$, integrate k times, no constants: $\frac{xe^x\cos x}{(D-1)^2} = e^x(2\sin x - x\cos x)$ (slide 006·9 is wrong). ESA: $\frac{x^2e^{-x}+\sin^2 x}{D^3-2D^2+D} = e^{-x}(-\frac{x^2}{4} - x - \frac{11}{8}) + \frac{x}{2} - \frac{\cos 2x}{25} + \frac{3\sin 2x}{100}$.

3 · The Type 5 rule: $X = xV$

Type 5: $X = xV$. $\frac{1}{f(D)}(xV) = x\frac{1}{f(D)}V - \frac{f'(D)}{(f(D))^2}V$, both parts by Type 2. CW P19: $\frac{x\sin 3x}{D^2-1} = -\frac{x\sin 3x}{10} - \frac{3\cos 3x}{50}$. CW P20: $\frac{x\cos 2x}{D^2-5D+6} = \frac{x(\cos 2x-5\sin 2x)}{52} - \frac{73\sin 2x}{1352} - \frac{5\cos 2x}{169}$. Check: $x\sin ax = \mathrm{Im}(xe^{iax})$, then Type 4 with $V = x$.

2 Keyword index

17 terms; the number in brackets is the topic.

$\frac{1}{D^k}$ (2). Integrate k times, no constants.

$\sin^2 x$ (2). $\frac{1-\cos 2x}{2}$.

$\mathrm{Im}(xe^{iax})$ (3). $x \sin ax$: Type 4 with $a \to ia$ (check only).

a (1). Coefficient of x in the exponent.

$D \to D + a$ (1). Replace every D in $f(D)$.

$f'(D)$ (3). Derivative of $f(D)$ with respect to D.

$V = \sin bx,\ \cos bx$ (1). After the shift, Type 2 with $D^2 \to -b^2$.

$x^m e^{ax}$ (2). After the shift, Type 3 series up to D^m.

Correction term (3). $\frac{f'(D)}{(f(D))^2}V$, subtracted.

Exponential shift (1). $\frac{1}{f(D)}e^{ax}V = e^{ax}\frac{1}{f(D+a)}V$.

Failure by shift (1). AE root a gives a factor D in $f(D + a)$.

Integration by parts (2). $\int x \cos x\, dx = x \sin x + \cos x$.

Shifted operator (1). $f(D + a)$, expanded and simplified.

Superposition (2). PI = PI_1+PI_2.

Type 4 (1). $X = e^{ax}V$, V a function of x.

Type 5 (3). $X = xV$.

Type 5 rule (3). $\left[\frac{x}{f(D)} - \frac{f'(D)}{(f(D))^2}\right]V$.

3 Rapid recall

1	Type 4 rule?	$\frac{1}{f(D)}(e^{ax}V) = e^{ax}\frac{1}{f(D+a)}V$.
2	Why $D + a$?	$D(e^{ax}V) = e^{ax}(D+a)V$ (product rule).
3	$(D+1)^2 - 2(D+1) + 4$?	$D^2 + 3$.
4	$\frac{e^x \cos x}{D^2-2D+4}$?	$e^x \frac{\cos x}{D^2+3} = \frac{e^x \cos x}{2}$.
5	$\frac{e^x \sin x}{(D-1)^2}$?	$e^x \frac{\sin x}{D^2} = -e^x \sin x$.
6	$\frac{e^{2x} \cos x}{D^2+4D+3}$ (CW P17)?	$\frac{e^{2x}(7\cos x + 4\sin x)}{130}$.
7	$\frac{x^2 e^{3x}}{D^2+2}$?	$\frac{e^{3x}}{11}(x^2 - \frac{12x}{11} + \frac{50}{121})$.
8	$\frac{xe^x \cos x}{(D-1)^2}$?	$e^x(2\sin x - x\cos x)$; slide 006·9 is wrong.
9	$\frac{xe^{3x}}{D^2-3D+2}$?	$\frac{e^{3x}}{2}(x - \frac{3}{2})$.
10	$\frac{x^2 e^{-x}}{D^3-2D^2+D}$?	$e^{-x}(-\frac{x^2}{4} - x - \frac{11}{8})$.
11	PI of $\sin^2 x$ for $D^3 - 2D^2 + D$?	$\frac{x}{2} - \frac{\cos 2x}{25} + \frac{3\sin 2x}{100}$.
12	$\frac{e^{2x}}{(D-2)^2}$ by the shift?	$e^{2x}\frac{1}{D^2}1 = \frac{x^2 e^{2x}}{2}$.
13	Type 5 rule?	$\frac{1}{f(D)}(xV) = \left[\frac{x}{f(D)} - \frac{f'(D)}{(f(D))^2}\right]V$.
14	$\frac{x \sin 3x}{D^2-1}$ (CW P19)?	$-\frac{x\sin 3x}{10} - \frac{3\cos 3x}{50}$.
15	$\frac{x \cos 2x}{D^2-5D+6}$ (CW P20)?	$\frac{x(\cos 2x - 5\sin 2x)}{52} - \frac{73\sin 2x}{1352} - \frac{5\cos 2x}{169}$.
16	Which ESA question uses Type 4?	Jan-24 set 3 2c, $(D^3 - 2D^2 + D)y = x^2 e^{-x} + \sin^2 x$ (= CW P18).

Mnemonic (study aid). *"Exponent out, D moves up by a, then V decides."* The three steps of Type 4 (study aid).

4 Last 10 minutes before the exam

Top facts

1. Type 4: $\frac{1}{f(D)}(e^{ax}V) = e^{ax}\frac{1}{f(D+a)}V$: e^{ax} out to the left, every $D \to D+a$, simplify, then the rule for V.
2. After the shift: $\sin/\cos \to D^2 \to -b^2$; polynomial → series up to D^m; D^k alone → integrate k times, no constants.
3. An AE root a needs no failure rule: $f(D+a)$ has a factor D ($\frac{e^x \sin x}{(D-1)^2} = -e^x \sin x$).
4. Type 5: $\frac{1}{f(D)}(xV) = x\frac{1}{f(D)}V - \frac{f'(D)}{(f(D))^2}V$, both by Type 2.
5. ESA Jan-24 s3 2c (= CW P18): PI $= e^{-x}(-\frac{x^2}{4} - x - \frac{11}{8}) + \frac{x}{2} - \frac{\cos 2x}{25} + \frac{3\sin 2x}{100}$.
6. CW P19: $-\frac{x\sin 3x}{10} - \frac{3\cos 3x}{50}$; CW P20: $\frac{x(\cos 2x - 5\sin 2x)}{52} - \frac{73\sin 2x}{1352} - \frac{5\cos 2x}{169}$.
7. Slide slips: 006·9 should give $e^x(2\sin x - x\cos x)$; 006·5's CF needs $c_2 \sin\sqrt{3}x$.

Top formulas

E1.1 · Type 4: the exponential shift

ESA ★★

$$\frac{1}{f(D)}\left(e^{ax}V\right) = e^{ax}\,\frac{1}{f(D+a)}\,V = e^{ax}\left[f(D+a)\right]^{-1}V$$

GIVES The PI of e^{ax} times any function V.

APPLIES $X = e^{ax}V$; a = coefficient of x in the exponent.

NOTE Write e^{ax} to the left first, then replace every D by $D+a$; V is untouched.

S006·4 · S006·10 · T006 00:00:52 · T006 00:01:00 · T006b 00:10:19 · ESA Jan-24 s3 2c

E1.3 · Then $V = \sin bx$ or $\cos bx$ (Type 2)

ESA ★★

$$e^{ax}\frac{1}{f(D+a)}\cos bx: \quad D^2 \to -b^2 \text{ in } f(D+a)$$

GIVES The second half of the work when V is trigonometric.

APPLIES After the shift; rationalise a leftover D (class 004).

NOTE Use b of V, not a of the exponential: $e^x\cos x$: $D^2 \to -1$.

S006·6 · T006 00:04:20 · T006b 00:10:53 · CW U2 P17

E2.1 · Then $V = x^m$ (Type 3)

ESA ★★

$$\frac{1}{f(D)}\left(x^m e^{ax}\right) = \frac{e^{ax}}{b_0}\left[1 \pm \varphi(D)\right]^{-1} x^m, \quad f(D+a) = b_0\left[1 \pm \varphi(D)\right]$$

GIVES The PI of a polynomial times an exponential.

APPLIES After the shift, $f(D+a)$ has a constant term $b_0 = f(a) \neq 0$.

NOTE Expand up to D^m only (class 005 series). The PI is e^{ax} times a polynomial of degree m.

S006·11 · S006·12 · T006b 00:12:47 · T006b 00:13:06 · T006 00:12:09 · CW U2 P18

E2.2 · When the shift leaves only D^k

ESA ★★

$$\frac{1}{(D-a)^k}\left(e^{ax}V\right) = e^{ax}\frac{1}{D^k}V = e^{ax}\underbrace{\int\cdots\int}_{k} V\,dx\cdots dx$$

GIVES The PI by k integrations (no constants).

APPLIES $f(D) = (D-a)^k$, e.g. $D^2 - 2D + 1 = (D-1)^2$ with $a = 1$.

NOTE Integrate by parts for $x\cos x$; "without writing the constant of integration, that is very important".

S006·8 · S006·9 · S006·14 · T006 00:06:40 · T006 00:07:16 · T006b 00:14:42

E3.1 · Type 5: $X = xV$

ESA ★★

$$\frac{1}{f(D)}(xV) = \left[\frac{x}{f(D)} - \frac{f'(D)}{\left(f(D)\right)^2}\right]V = x\,\frac{1}{f(D)}V - \frac{f'(D)}{\left(f(D)\right)^2}V$$

GIVES The PI of x times a function (usually $x\sin ax$, $x\cos ax$).

APPLIES $X = xV$; $f'(D) = \frac{d}{dD}f(D)$.

NOTE $\frac{x}{f(D)}V$ means: apply $\frac{1}{f(D)}$ to V, then multiply by x.

S006·4 · T006 00:01:28 · CW U2 P19 · CW U2 P20